User Stats | Question Stats | Question Details

Question Details

All Attempts | Export to CSV | Export to Excel

Has Start Date 22/08/2025

Has End Date 29/08/2025 Apply

All Attempts (863) What do the statistics on this page mean?

Question 1 Difficulty: 1

What is the CPU - Memory gap?

The distance between the CPU and memory modules	16	(1.85 %)
➡ The difference in processing speed between a CPU and memory	800	(92.27 %)
Memory stores data, whereas the CPU processes it	3	(0.35 %)
The difference in storage capability between main memory and cache on the CPU	35	(4.04 %)
No Response	13	(1.5 %)

Average Grade: 0.93 / 1 (92.7 %)
Standard Deviation: 26.01 %
Point Biserial: 0.42
Discrimination Index: 21.89 %

Question 2 Difficulty: 1

Java objects in particular are problematic in data oriented design. This is mainly due to the fact that Java objects live on the heap. What is the problem with Java objects? Select all that apply.

➡ Heap objects can only be accessed indirectly and	790	(91.54 %)

these indirect access are hard to predict

- Heap objects are distributed non-sequentially throughout memory — 806 (93.4 %)
- A heap is a tree structure which must be traversed, adding to access times — 100 (11.59 %)
- Java objects do not support compile time polymophism (i.e.: no early binding) — 291 (33.72 %)
- Java objects do not support run-time polymorphism (i.e: no late binding) — 41 (4.75 %)
- No Response — 30 (3.48 %)

Average Grade: 0.62 / 1 (62.46 %)
Standard Deviation: 28.75 %
Point Biserial: n/a
Discrimination Index: 37.00 %

Question 3 Difficulty: 1

Why is C++ such a good fit for data oriented design? Select all that apply.

- It is a multi-paradigm language and thus supports 'everything and the kitchensink'. — 269 (31.17 %)
- The default object layout minimizes indirection — 748 (86.67 %)
- It has extensive

Average Grade: 0.49 / 1 (49.13 %)

support for run-time polymorphism, through the use of virtual and abstract methods. 165 (19.12 %)

Standard Deviation: 32.06 %
Point Biserial: n/a
Discrimination Index: 43.35 %

➡ It has extensive support for compile-time polymorphism, using for example templates. 743 (86.1 %)

No Response 29 (3.36 %)

Question 4 Difficulty: 1

What is CPU pipelining?

Running your code through multiple CPUs in parallel 9 (1.03 %)

➡ The CPU processes instructions using overlapping stages 834 (95.64 %)

The data flows between memory and the CPU through different buffers called 'caches' 4 (0.46 %)

Average Grade: 0.97 / 1 (96.64 %)
Standard Deviation: 18.02 %
Point Biserial: 0.37
Discrimination Index: 10.30 %

A CPU can only process instructions in the order in which it receives them 6 (0.69 %)

No Response 19 (2.18 %)

Question 5 Difficulty: 1

What is defensive programming?

Programming applications for military use	0	(0 %)
➡ Programming habits that reduce the chance of errors	823	(94.38 %)
Checks to find errors, such as unit testing and static analysis	28	(3.21 %)
Programming using a 'safe' language, such as Rust or Java	1	(0.11 %)
No Response	20	(2.29 %)

Average Grade: 0.95 / 1 (95.37 %)
Standard Deviation: 21.02 %
Point Biserial: 0.38
Discrimination Index: 13.30 %

Question 6 Difficulty: 1

Why does the CS department insist on unit tests? Select all that apply.

➡ Good unit tests require a testable class hierarchy, resulting in a better design	691	(80.07 %)
➡ Unit tests allow programmers to catch bugs early, when they are easy to fix	809	(93.74 %)
The department loves busy work	13	(1.51 %)
➡ Full unit test coverage allows the programmer to catch bugs that are hard	670	(77.64 %)

Average Grade: 0.76 / 1 (75.94 %)
Standard Deviation: 31.43 %
Point Biserial: n/a
Discrimination Index: 42.66 %

to catch in any other way

Unit tests facilitate regression testing — 730 (84.59 %)

No Response — 36 (4.17 %)

Question 7 Difficulty: 1

Why is security different? Choose the best answer.

Because it must be solved in the early phases at requirement time — 731 (83.73 %)

It isn't, it's just another quality aspect — 38 (4.35 %)

Because it is complicated — 13 (1.49 %)

Because you cannot test for it — 68 (7.79 %)

No Response — 23 (2.63 %)

Average Grade: 0.85 / 1 (84.7 %)
Standard Deviation: 35.99 %
Point Biserial: 0.49
Discrimination Index: 46.78 %

Question 8 Difficulty: 1

What does "shift left" mean? Choose the best answer

A renewed focus on software quality aspects, esp. security — 781 (89.36 %)

a << b, a way to multiply numbers by powers of 2 — 30 (3.43 %)

A focus on software performance, esp. in the context of C++ — 10 (1.14 %)

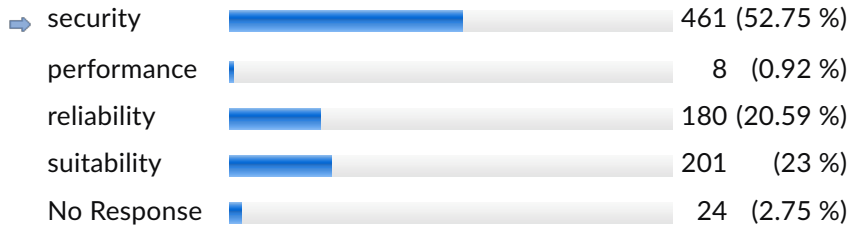
A move towards higher developer productivity — 30 (3.43 %)

No Response — 23 (2.63 %)

Average Grade: 0.9 / 1 (90.5 %)
Standard Deviation: 29.32 %
Point Biserial: 0.44
Discrimination Index: 28.33 %

Question 9 Difficulty: 1

What should be the number #1 priority in software development



Average Grade: 0.53 / 1 (53.42 %)
Standard Deviation: 49.88 %
Point Biserial: 0.46
Discrimination Index: 72.53 %